Diagnosing Parkinson's By Applying Graph Neural Networks to MRI

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Introduction

Parkinson's is a debilitating neurodegenerative disorder; it is the second-most common neurodegenerative disease in the elderly, and is primarily associated with degeneration of dopamine-producing neurons in the midbrain. It is difficult to diagnose, primarily relying upon physical assessment. However, another potential biomarker is regional thinning of the cortical surfaces. By extracting surfaces from MRI scans and converting them to graph representation, we may be able to diagnose Parkinson's by applying graph-based neural networks to cortical surfaces

Methods

The project relies upon graph-based neural networks with graph convolutional layers, which sum up average features of neighbors before applying weight matrix and transfer function. Another variant is graph isomorphic layers, which sum up features before feeding through MLP and then applying transfer function.

The data is sourced from Parkinson's Progression Markers Initiative, and surfaces were extracted using the FreeSurfer software. The geometric mesh was converted into a graph, with various features (curvature, thickness, etc.) considered, and edges linking points (nodes) in the same triangular face. Figures 1-3 provide visualizations of the pipeline.

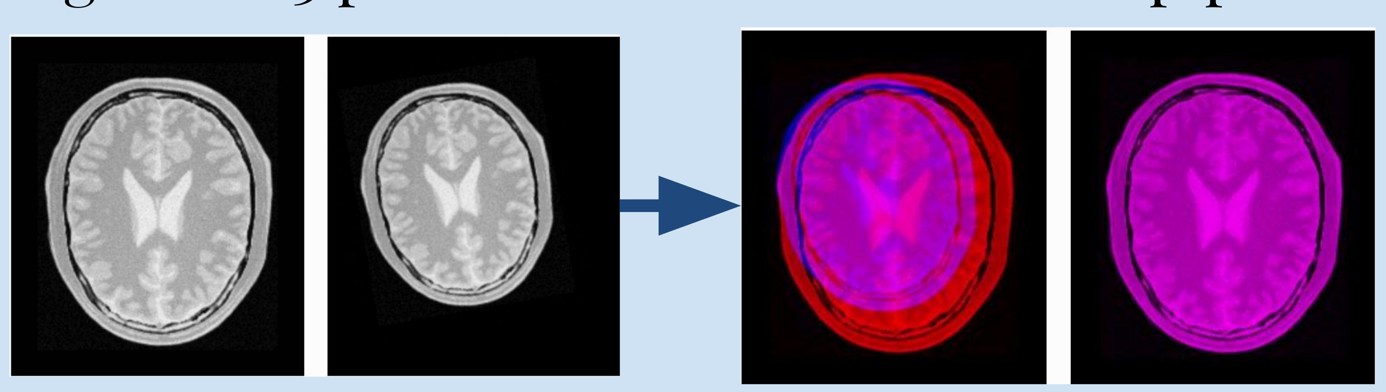

Figure 1: Registers to known space ("atlas") w/ affine transformation

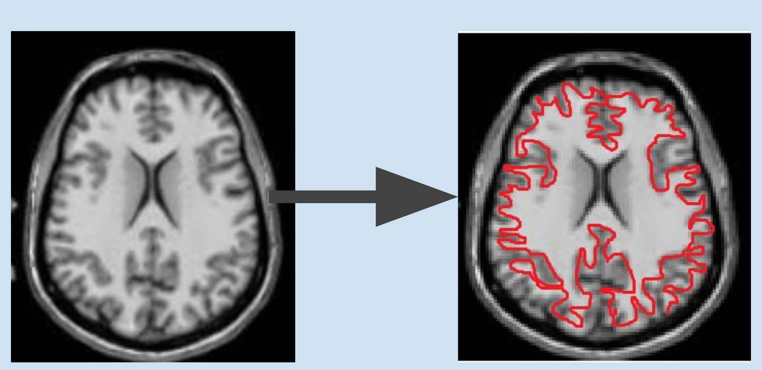

Figure 2: Classifies points as white matter/gray matter

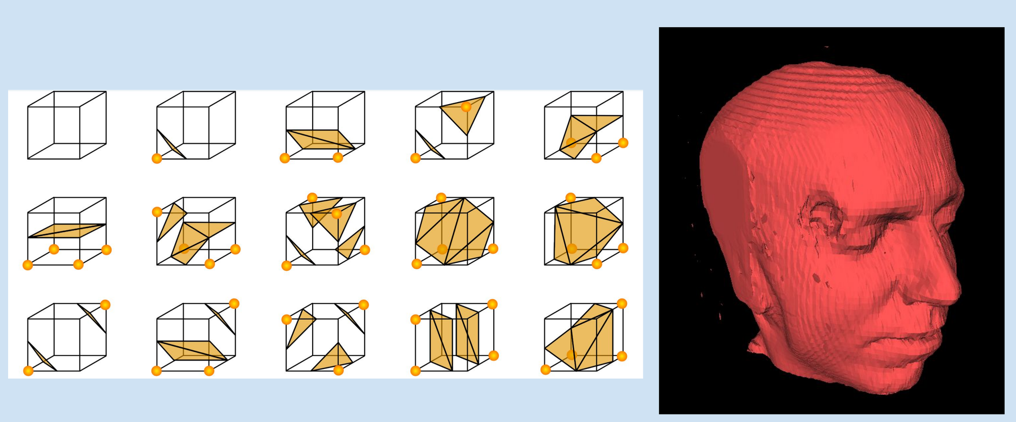

Figure 3: Determines geometric mesh through Marching Cubes algorithm

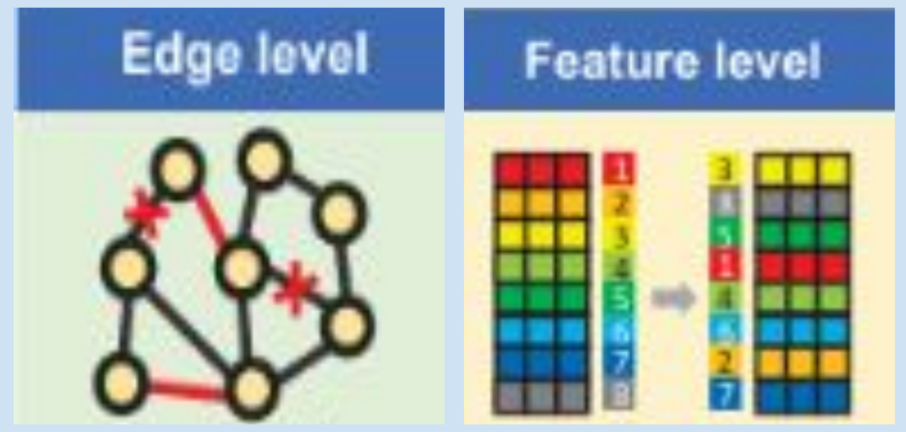

Figure 4: Data Augmentation

Figure 4 displays the augmentation processes used; cutting and rewiring a small proportion of the edges and adding random noise to the node features.

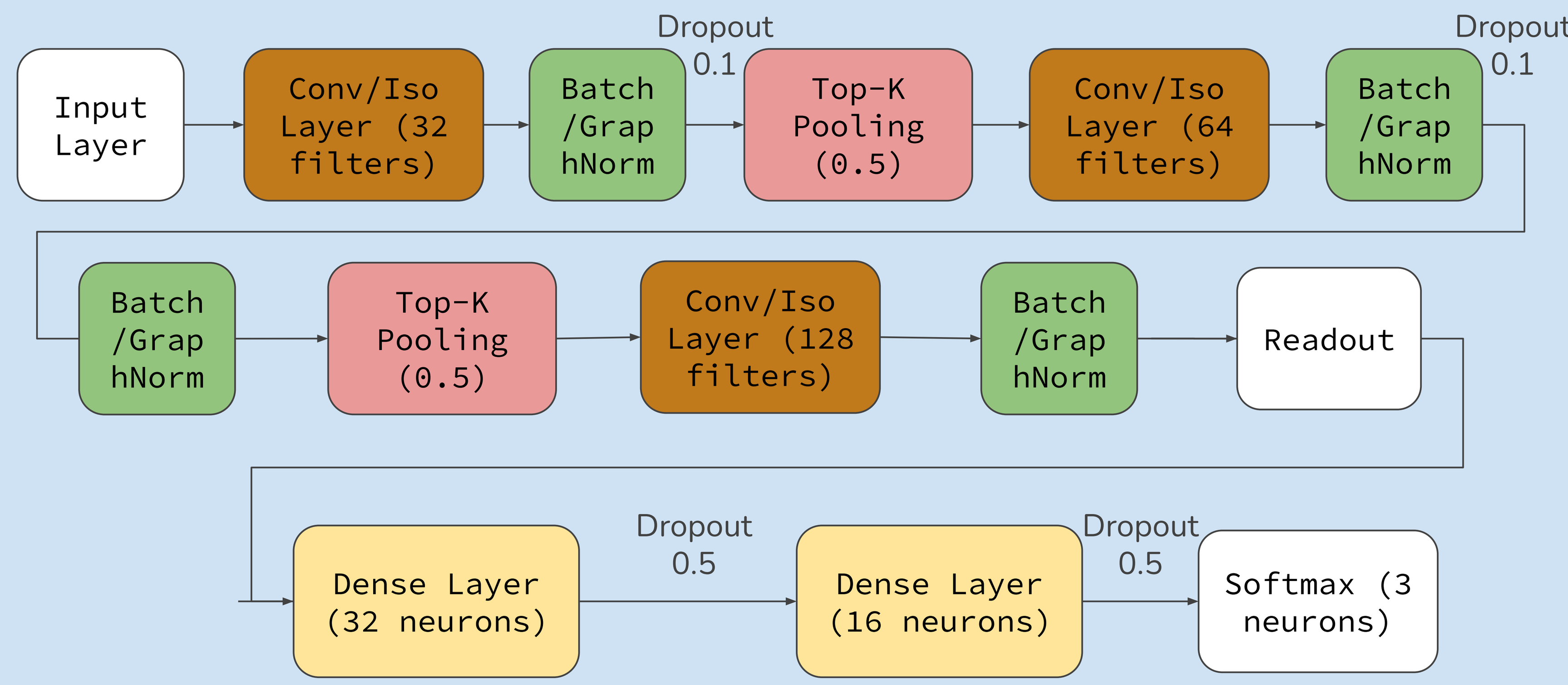

Figure 5: Model Architecture

Each model used had the architecture displayed in Figure 5. For the Top-K pooling, each node is scored based on its features and a trainable weight-vector. Then the top-K nodes are retained and remaining node features are scaled based on their score. Models used are described below.

Model 1	Model 2	Model 3
GCN & Batch normalization	GIN & Graph normalization	GCN & Graph Normalization w/ LogReg Model

Results

A logistic regression model was trained on patient evaluations, achieving 81.8% testing accuracy. This model was combined with a Graph-based NN.

Model 3: 84% Testing Accuracy

Predicted⤔ Actual ⤈	Control	Prodromal	PD
Control	154	23	12
Prodromal	11	162	15
PD	7	20	155

Table 1: Combined Confusion Matrix

Model 1: 56% Testing Accuracy

Predicted⤔ Actual ⤈	Control	Prodromal	PD
Control	151	38	0
Prodromal	22	166	0
PD	93	89	0

Table 2: GCN Confusion Matrix

Model 2: 55% Testing Accuracy

Predicted⤔ Actual ⤈	Control	Prodromal	PD
Control	113	61	15
Prodromal	14	167	7
PD	57	97	28

Table 3: GIN Confusion Matrix

Conclusion

The graph-based neural networks showed some ability to distinguish stages of Parkinson's Disease, particularly in the intermediate stage (Table 2 & 3). It improved overall performance when combined with logistic regression l (Table 1), but was insufficient as a standalone model, heavily underpredicting the PD class (Table 2 & 3).

The model contains only ~26,000 parameters, compared to 8–41 million in prior work. Hardware and GPU driver limitations prevented GPU acceleration, restricting model size and likely reducing its ability to capture complex graph patterns.

In the future, we intend on applying pre-trained 3d CNN to compare against the graph-based NN. We also plan on tuning the Graph Neural Networks via increasing the size of the network, experimenting with different graph constructions and representations and more advanced fusion methods.